Visual Analytics of Character Dynamics and Sentiment Evolution: A Case Study of the *Friends* TV Series

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Abstract—This paper presents a visual analytics approach to analysing the scripts of the *Friends* TV series, focusing on uncovering latent insights through transcript data and advanced visualization techniques. The analysis aims to explore character personas by identifying traits, behaviours, and emotional tendencies, while also examining the sentiment dynamics to map emotional trajectories across episodes and seasons. The study further investigates the evolution of character interactions, revealing how relationships develop over time and contribute to the narrative structure. By integrating sentiment analysis, network visualization, and narrative analytics, the research highlights key trends and pivotal moments that define the series' emotional and relational dynamics. The findings demonstrate the potential of visual analytics as an effective tool for analysing narrative-driven media, offering fresh perspectives on storytelling and providing a framework that can be applied to other television series and similar contexts.

1 Problem Statement

The television series "Friends" has been extensively analysed for its cultural impact and character development. However, there is a lack of comprehensive studies employing visual analytics to examine character dynamics and sentiment evolution throughout the series. Existing research has explored aspects such as conversational networks in TV series [1] and the use of natural language processing (NLP) and visualization for analysing fictional characters [2]. Yet, these studies do not specifically focus on "Friends" or integrate multiple analytical methods to provide a holistic view of character interactions and emotional trajectories.

This research aims to fill this gap by applying a visual analytics approach to the "Friends" TV series, focusing on:

1. Character Persona Analysis: Identifying traits, behaviours, and emotional tendencies of main characters to understand their development over time.
2. Sentiment Dynamics: Mapping emotional trajectories across episodes and seasons to uncover patterns and pivotal moments.
3. Evolution of Character Interactions: Examining how relationships among characters develop and influence the narrative structure.

By integrating sentiment analysis, network visualization, and narrative analytics, this study seeks to provide fresh perspectives on storytelling within "Friends" and demonstrate the effectiveness of visual analytics in analysing narrative-driven media. The findings could offer a framework applicable to other television series and similar contexts, contributing to the broader field of media analysis.

2 State of the Art

The dataset under study—TV show transcripts—falls within text analysis, focusing on character dynamics and sentiment analysis in *Friends*. Several prior works provide valuable insights for this project.

Krstajić et al. (2011)[3] introduced *CloudLines*, a visualization method for summarizing event episodes across multiple time series. Events are represented as semi-transparent circles, with their size and opacity indicating importance. This approach, published in IEEE InfoVis, inspired our visualization of emotionally significant moments in *Friends*, where key emotional peaks are visually highlighted. This method's compact design suits our need to condense a decade-long series into a digestible format.

Mintz et al. (1997)[4] proposed tile maps for visualizing temporal trends using calendar layouts. Data values are encoded by tile lightness, enabling intuitive interpretation. While this older work does not stem from a top-tier journal, its foundational concept was critical in visualizing episode ratings and viewership trends across seasons. By mapping data to a calendar-like format, we effectively captured fluctuations in audience engagement, which are comparable to temporal event patterns in Mintz et al.'s study.

Alper et al. (2011)[5] introduced *LineSets*, a method for visualizing overlapping relationships in networks. Published in IEEE Transactions on Visualization and Computer Graphics, their work was instrumental in mapping character interactions in *Friends*. LineSets provided an intuitive depiction of character dynamics, showing how interactions evolved over time and contributed to the broader narrative. This approach directly aligns with our goal of analyzing relational structures among the core six characters and their network.

Zahiri (2017)[6] applied sequence-based convolutional neural networks (CNNs) for emotion detection in TV show transcripts, focusing on nuanced sentiment extraction from dialogues. While not from a visualization-focused journal, this work is critical to our study as it demonstrates the utility of deep learning for uncovering emotional patterns. We incorporated a similar CNN model to map emotional trajectories, complementing our visual analytics with computational precision.

These studies demonstrate the synergy between advanced visualizations and sentiment analysis, showcasing their relevance to analyzing narrative-driven media like *Friends*. Krstajić et al.'s focus on summarizing events through salience-based markers aligns well with our need to distill emotional peaks from lengthy transcripts. Mintz et al.'s tile maps, though simplistic, offer a structured way to analyze temporal patterns, bridging raw data with intuitive visualization. Alper et al.'s work on network visualization addresses relational insights, which are critical for understanding character dynamics. Finally, Zahiri's approach highlights how machine learning can extract actionable insights from unstructured narrative data, aiding in detecting emotional subtleties that traditional visualization may miss.

By combining these methods, our study integrates sentiment analysis with visual analytics, enabling a comprehensive exploration of character interactions, emotional patterns, and audience reception in *Friends*. The adaptability of these techniques offers a robust framework for similar studies in narrative-driven datasets.

3 Properties of the Data

We used three csv files for this analysis first one containing all the ten seasons transcript of friends as well as the speakers name arrange in a chronological order using season, episode, scene and utterance order these three columns are also present in the other two files and therefore was used for merging them when needed . The second one containing the meta data like air date, IMDb rating , views for each episode as well as the directors and writers names, the last file contains sentiment data based on Zahiri's work (2017) for the first four seasons of friends. The data originates from the Character Mining repository which includes references to scientific explorations using this data.

Figure 1. pairplot of emotions feature of the first four seasons of friends

(Figure 1.) shows us the pairplot of the feature emotion which is based on Zahiri (2017)[6] work. We can see the trend in the change of emotion as seasons change for instance the emotion of joy becomes more prominent and, in the contrary, the joyful emotion becomes less prominent as scenes changes.

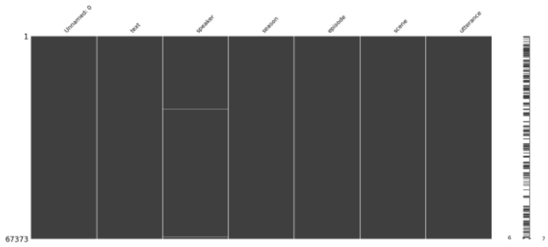

Figure 2. missingno graph showning missing values (horizontal white lines)

4. Analysis

4.1 Approach and plan

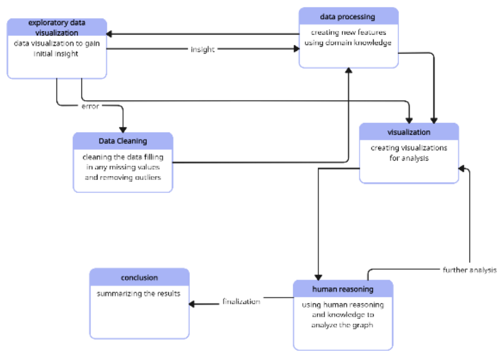

Diagram 1. Our Analysis workflow plan

We first checked the data for any missing values using the visual analysis tool called missingno[7] , its provides a small toolset of flexible and easy-to-use missing data visualizations and utilities that allows you to get a quick visual summary of the completeness (or lack thereof) of your dataset. On first look it seemed that the speaker column had some missing values but on further inspection we found that it was just scene directions, so our data had no missing values. Our scope of the research does not get effected by outliers as our focus of this analysis is on the transcript and the sentiment, so we ignored them.

After the data cleaning we did some data preparation , we calculated sentiment score using VADER[8] setiment analyzer we use it because its a effective for analyzing sentiments expressed in social media and dialogues.

After we where done with feature engeenearing we start our visual analysis with the focus on character wise persona analysis , sentiment dynamic analysis combined with analysis or overall plot and characted evolution.

We plan on doing the following:

- We are going to create a network graph using the pair wise sentiment score we calculated before hand to study the relation between the characters and how it develops as show goes on.
- We then plan on observing the overall sentiment presented by the characters at different parts of the scenes
- We also plan on looking at the in general pattern of sentiment expressed throught the different temporal scales.

- We will use spider wed graph to illustrate the the contrast in the sentiment used by the characters and also the writers story style.
- Story flow visualization of character screen time and big group meetup in the story.
- Heatmap visualizing the rating and views of the show .
- Comparing character popularity using boxplot
- Networkgraph of the main crew of the show to visualizeco-worker relation

4.2 Process

In this section we did visual analysis with aim to achieve the following task :

1. Evolution of Character Interactions: Examining how relationships among characters develop and influence the narrative structure.
2. Analyse character traits as well as the author writing style.
3. Gaining insights about the emotional trajectories across episodes and seasons to uncover patterns and pivotal moments.
4. Visualize the performance of the show.

Task 1

First of all we calculated sentiment score using VADER setiment analyzer, we use it because its a effective for analyzing sentiments expressed in social media and dialogues. In order to perform sentiment analysis using VADER we first clear speaker data by removing the scene direction rows , removing all the punctuation and numbers as well as stopwords for the data.

After calculating sentiment score we also calculated the compound sentiment score which provides a single metric for sentiment polarity. Which we are going to use for our characters relation visualisation. Here I need to mention that we didn't have the data about which line is directed towards another character , so we just calculated aggregate sentiment score for each character pair my taking there mean compound score per scene and speaker so basically we presume that the sentiment is directed towards every body who is present at the scene and then we calculated pairwise compound score for each pair using the average coumpound score of the characters in the pair. We use this data to create a network graph for friends characters. (Figure 3.)

(Figure 3.) combines network graph together with pair wise sentiment we calculated to show the intensity and nature of characters interaction, the green hue shows positive interaction where as the red hue shows negative interactions. The size of the circle shows the amount of dialogue the character had. Form the graphs we can observe how relationship between characters changes with change in seasons.

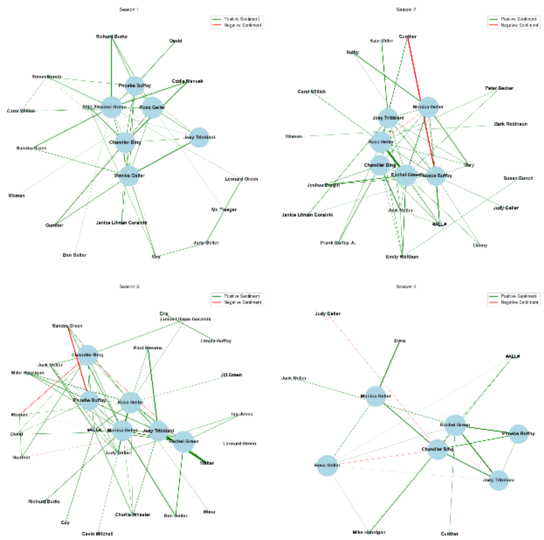

Figure 3. sentiment analysis network

Besides this we also used plotly.js library to create a network graph of the top characters , directors and writers to get insight about the relation between them (figure 4.). Here we have used spring layout algorithm which is force-direction layout algorithm that simulates physical forces acting on the nodes of graph.

Attractive forces: pull connected nodes closer together.

Repulsive forces: push all nodes apart to prevent overlap.

The goal is to minimize energy in the system, which results in a visually intuitive representation where:

- Nodes with many shared connections (highly connected or "central" nodes) are placed closer together.
- Nodes with fewer or weaker connections are positioned farther apart.

This insight we can get from this graph are as follows:

- Nodes that are closer together tend to have stronger relationships (e.g., characters who frequently share scenes, or a director and characters who worked together often).
- Nodes that are farther apart are less connected or not directly related.
- Groups of nodes forming clusters indicate tightly knit subgroups within the network, such as:
- A group of characters who frequently interact in scenes.
- A writer or director heavily associated with specific characters or episodes.
- Outliers:Nodes positioned far from the center or clusters are typically outliers, representing nodes with minimal or no significant relationships.

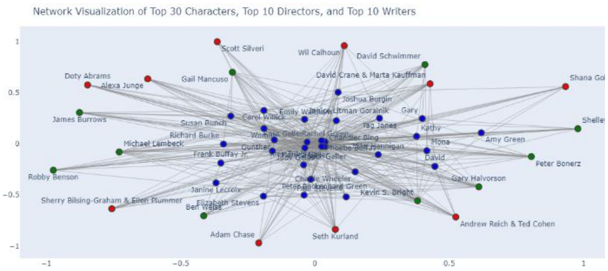

Figure 4. co-worker network using plotly.js

Task 2

For this task we decided to use spider wed chart as it's a good way of comparison of categories with scaler values. Here we are going to use the frequency of sentment which has 7 categories namely: mad, neutral , joyful, scared, sad, powerful and peaceful. This was already present in the data that we got for kaggle.

So we use it to compare the nature of sentiment of the main characters of the show friends and also to see the contrast between the sentiment present in the top ten writers of the script, we only compared the top characters and author for the cleartity of visualisation as well as there significance.

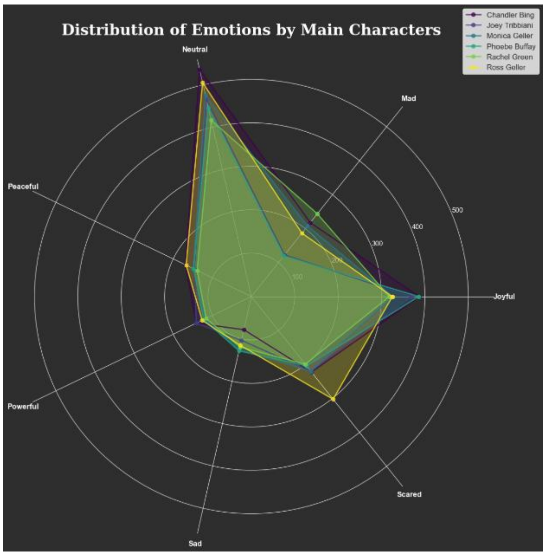

Figure 5. (a) spider web graph of main characters sentiment.

It shoes the frequency of each type of emotions expressend by them throuhgt the first four seasons. It's a great a way to compare there persona.

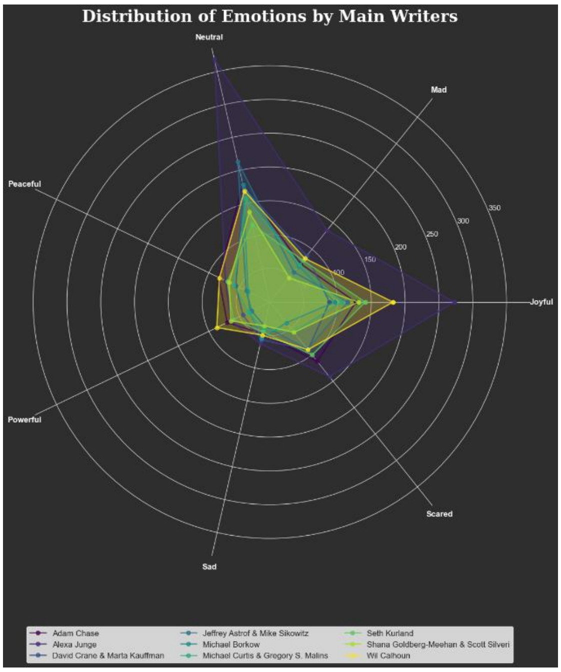

Figure 5. (b) spider web graph of the top writers by amount of transcript written.

(Figure 5.) (a) gives information about the personality of the character and how they compare with the rest of the main characters. Whereas figure 5. (b) gives information about the writing style of the authors and what kind of plot or emotions they like to potray.

Task 3

For this task we are going to use line graphs and a visualization approach similar to the CloudLines. Here we use the sentiment data we calculated using VADER .

(Figure 6.) this visualization provides us with information about the moments in every episode with the most positive and most negative dialogue. As well as the moment with the most positive and negative sentiment in the season. This gives us insight about the patterns in the drama, from the observation of the line graph it seems that there is a overlapping of the maximum and minimum sentiment lines in the episodes near the end of each seasion point at the presence of hyped of drama. Not only that its also noticible that majority of these red and green lines are near the end of the episode which may point towards presence of cliffhangers in end of the episode to keep the audience hooked in.

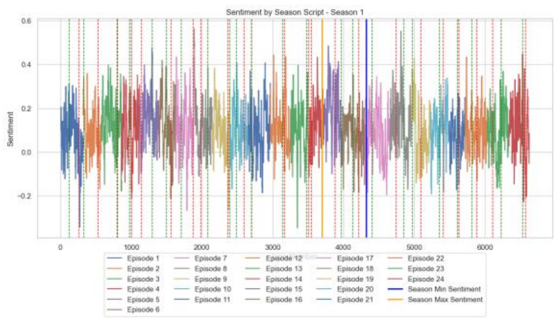

Figure 6. sentiment line graph

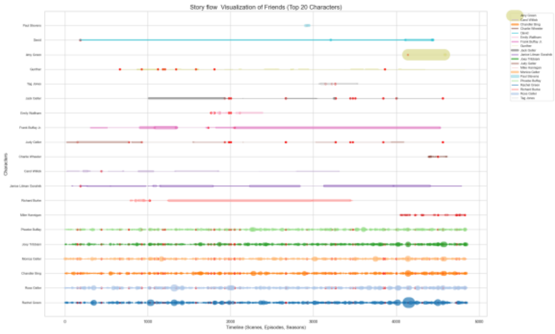

Figure 7. CloudLine graph of friends

(Figure 7.) this graph shows the timeline and the amount of dialogue the top 2o characters had in the show as well as the big gathering moments in the show. This graph gives us insight about the presence and importance of character at any point of the sceries, presuming that important characters get more screen time.

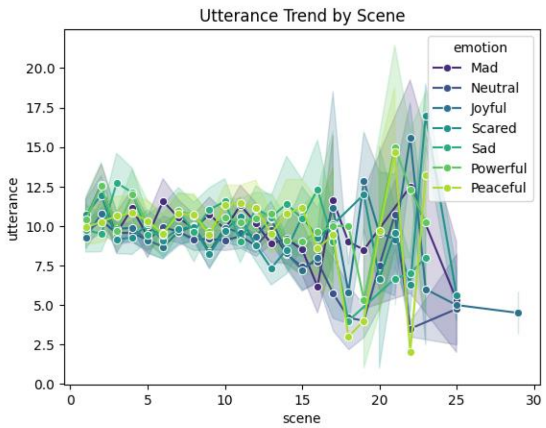

Figure 8. line plot of utterance by sentiment category

(Figure 8.) The graph clearly shows us the trend in fluctuation of emotion for the friend series. The ends scenes tends to have a lot of emotional drama to keep the audience hooked for the next episode. We can observe form the graph that the

Task 4

For this task we use heat map to visualise the IMDb rating and views for each epside per season.(figure 9.). The circle size represents the views where as the colour of the heatmap represents the rating. We can observe from the graph that the last two episodes of the friends series where the most popular both in terms of ration as well as views. The episodes with more views seems to also have better ratings on average.

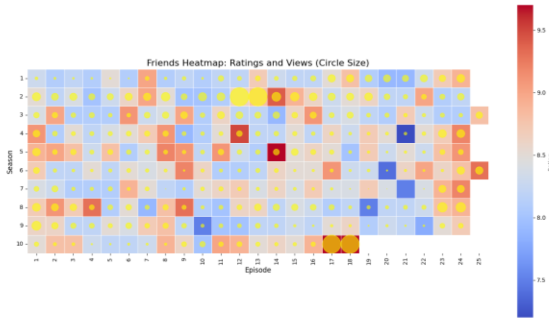

Figure 9. heatmap of rating and views of friends series

4.3 Results

Our analysis provides key insights into the character dynamics, emotional trajectories, and narrative structure of the TV series *Friends*:

Character Dynamics: The network graph (Figure 3) illustrates how main character relationships evolved, such as Ross and Rachel's fluctuating interactions reflecting their complex arc. Sentiment intensity, shown with color-coded edges, highlights pivotal moments. The co-worker network graph (Figure 4) shows close ties among directors, writers, and characters, emphasizing collaborative consistency and core production dynamics. (Figure 10) shows the importance of characters and there network coverage the more centered the point is the more connected it is in the network.

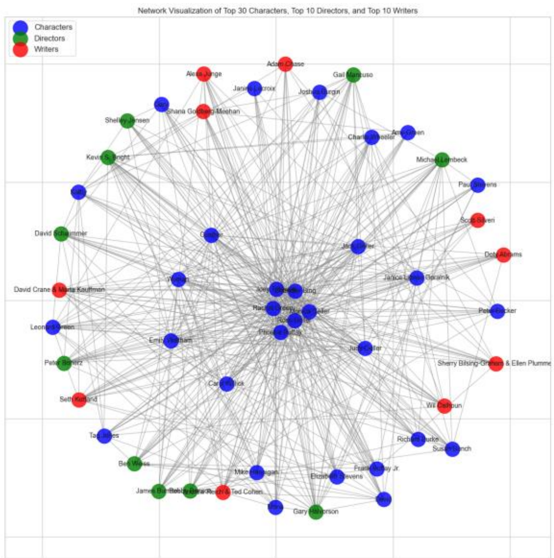

Figure 10 network graph (linesets)

Sentiment Dynamics: The sentiment line graph (Figure 6) identifies emotional peaks at season finales, coinciding with dramatic, cliffhanger moments. Overlapping sentiment extremes near season ends confirm heightened drama. Spider web charts (Figure 5) reveal emotional personas: Chandler's humor aligns with joyful sentiment, while Ross's blend of powerful and sad sentiments reflects his conflicted character arc.

Narrative Performance: The heatmap (Figure 9) links high episode ratings and viewership, with series finales performing best, validating effective storytelling. The CloudLine graph

(Figure 8) highlights emotional fluctuations in end scenes, often involving cliffhangers maintaining audience engagement between episodes.

5. Reflection

Reflection on the Approach and Process

The chosen approach effectively combined sentiment analysis, network visualization, and temporal trend mapping to answer our research questions. This integration of techniques allowed for a holistic exploration of the dataset. For instance, employing VADER for sentiment scoring and combining it with network graphs uncovered not just the nature of character interactions but also their evolution. Similarly, the spider web graphs provided a clear comparative framework for character personas and writing styles.

Limitations and Challenges:

The assumption that sentiment in a scene is directed toward all present characters oversimplifies interpersonal dynamics. Including explicit speaker-listener data would yield more precise results.Sentiment analysis relied heavily on VADER, which, while effective for social media and dialogues, may not capture subtleties like sarcasm or complex emotions inherent in *Friends*.Large, dense graphs, such as the co-worker network, were challenging to interpret without additional filtering or hierarchical views. Future work could explore interactive visualizations to address this.Sentiment data was limited to the first four seasons and even for the first four seasons not all the utterances where labled. Expanding this to all ten seasons would provide a more comprehensive view.

Employing advanced NLP models (e.g., transformer-based models like BERT) could improve sentiment accuracy by accounting for context and nuance. Also using interactive graphs like dynamic word cloud[9] , story line visualization [10] And SentiCompass[11] could have helped in providing a more indepth analysisUtilizing dynamic graph layouts or clustering algorithms could simplify dense network visualizations.Incorporating interactive tools, such as Plotly Dash, could enable users to explore specific subplots, characters, or episodes in more detail.The methodology can extend to other narrative-driven datasets, such as books or movie franchises. For broader applicability, adapting visualizations to highlight unique features (e.g., scene complexity or multi-plot structures) would be essential.The study successfully demonstrates how visual analytics can uncover intricate patterns in character dynamics, sentiment evolution, and narrative structures. While the approach has limitations, it offers a strong foundation for future analyses of media narratives. Analysts should prioritize refining sentiment precision, exploring interactive visualizations, and considering data-specific nuances to maximize insights.

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Table of word counts

Problem statement	216
State of the art	459
Properties of the data	187
Analysis: Approach	325
Analysis: Process	1114
Analysis: Results	200
Critical reflection	345